

BUILD OPS — working agreement

Construction ERP for Actuate Builders Inc. (ABI), a Philippine design-and-build, fit-out and MEP contractor. Next.js on Vercel, Supabase/Postgres.

Read `docs/PRD.md` before any schema, pricing or migration work. It holds the locked decisions, migration DDL, invariants and the WO-01→WO-18 sequence. It is too large to load automatically — open it when the task touches those areas. Where anything here and the PRD disagree, the PRD wins.

This is a refactor, not a rewrite

Three schema facts are verified against the live database and are not open:

1. Money is `BIGINT` centavos; percentages are integer basis points. No floats.
2. `tenant_id` is `NOT NULL` on all application tables, with RLS in place.
3. `bom_line_items.id` is a stable UUID already referenced by cost, budget, PO and RFQ records.

The relational spine exists. The defect is **grain and pricing**, not identity.

Hard constraints

- **Additive migrations only.** No `DROP COLUMN`, no `DROP TABLE`, no re-pointing an existing foreign key.
- **Never create a `scope_items` table or a `scope_item_id` column.** `bom_line_items` *is* the scope item (ADR-03). This is the most likely wrong instinct in this codebase and it costs a four-domain migration.
- **No general ledger work.** No journals, posting periods, cash accounts or reconciliation. ABI runs SAP; we build the interface, not a second book of record (ADR-07). The existing Finance module is frozen.
- **No float, double precision, unscaled numeric, or JS number in any monetary path.**
- Every new table: `tenant_id NOT NULL`, matching RLS policy, `created_at` / `updated_at` / `created_by`, audit participation.

- Every acceptance criterion ships as an automated test, not a manual check.
- Takeoff imports **upsert**, never delete-and-reinsert — that orphans downstream POs and vendor assignments.

Pricing model

Unit rates are **derived** from a DUPA (Detailed Unit Price Analysis), never typed. Material + labour (crew × hourly rate ÷ productivity) + equipment → Direct → OCM 800 bps + Profit 700 bps → VAT 1200 bps → Total → **Unit Rate**.

Intermediates compute unrounded; round half-up to centavos only at persistence and presentation. Full spec and the two canonical tests are in PRD §4.

Working style

- One work order per session. Confirm the plan before running any migration.
- Show migrations before running them, on staging first.
- State assumptions not covered by the PRD, and flag the expensive ones.
- If asked to do something that contradicts the constraints above, say so rather than complying.

Commands

```
[pnpm dev]           # dev server
[pnpm test]          # full suite — must pass before any work order is called done
[pnpm lint]
[supabase db diff]    # generate a migration
[supabase db push --dry-run] # ALWAYS dry-run before applying
```

Migrations run on staging first, never straight to production. Show me the SQL and the rollback before applying anywhere.

Ask before

- Applying any migration
- Adding a dependency
- Touching `supabase/migrations/` history

- Anything under `src/modules/finance/` (frozen — see ADR-07)